

IIAI30013

Computer Vision

HW2: Harris Corner Detection

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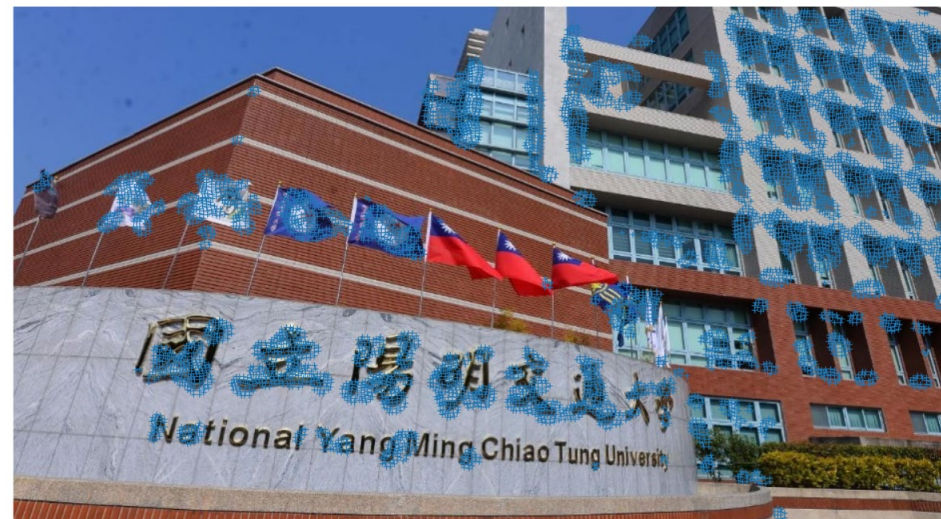
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HW2: Harris Corner Detection

- Homework due: 4/8 23:59
- Late submissions will incur a penalty of one point for each day overdue.
- The assignment allows a maximum extension of 3 days (it will not be accepted if submitted later than 3 days).
- Submit files: code and report (4 questions), and submit them in both **.zip** and **PDF** file formats respectively.
- This assignment can be carried out using [Colab](#) or completed on your PC.

HW2: Harris Corner Detection

- With the Harris corner detector described in slides (IIAI30013_05_Corner Detection.pdf), mark the detected corners on the image.



HW2: Harris Corner Detection

- **Algorithm**

- 1) Filter the image with a Gaussian.
- 2) Estimate intensity gradient in two perpendicular directions for each pixel.
- 3) Compute M matrix for each image window to get their *cornerness* scores.

$$M = \sum \begin{bmatrix} I_x^2 & I_x I_y \\ I_x I_y & I_y^2 \end{bmatrix} \quad R = \det(M) - k \cdot (\text{trace}(M))^2$$

- 4) Find points whose surrounding window gave large **corner response** ($f > \text{threshold}$)
- 5) Take the points of local maxima, i.e., perform non-maximum suppression

HW2: Harris Corner Detection

- Gaussian Smooth

gaussian blur of kernel size = 5



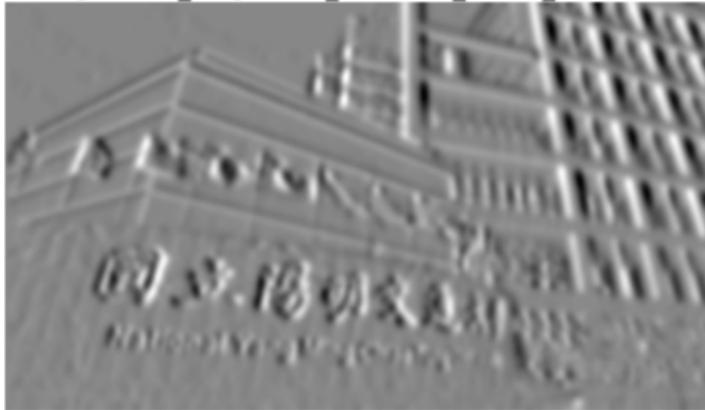
gaussian blur of kernel size = 10



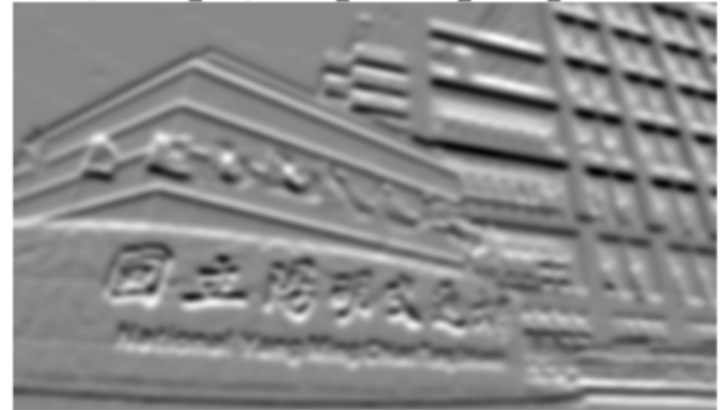
HW2: Harris Corner Detection

- Compute Gradients

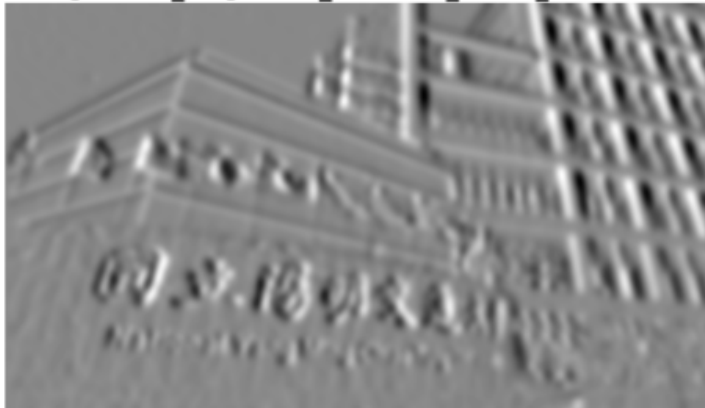
gradient_magnitude_Gaussian_kernel_size=5



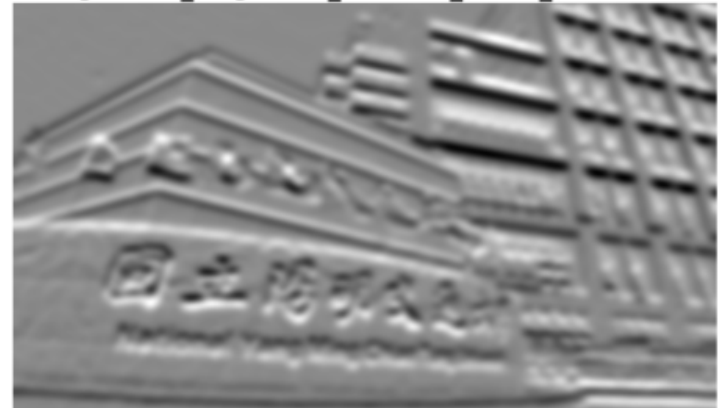
gradient_magnitude_direction_kernel_size=5



gradient_magnitude_Gaussian_kernel_size=10



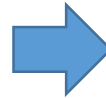
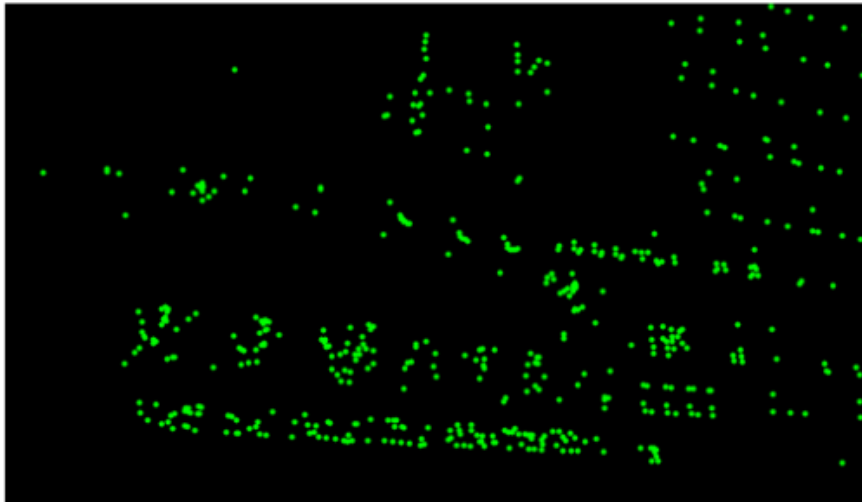
gradient_magnitude_direction_kernel_size=10



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- Non-Maximum Suppression

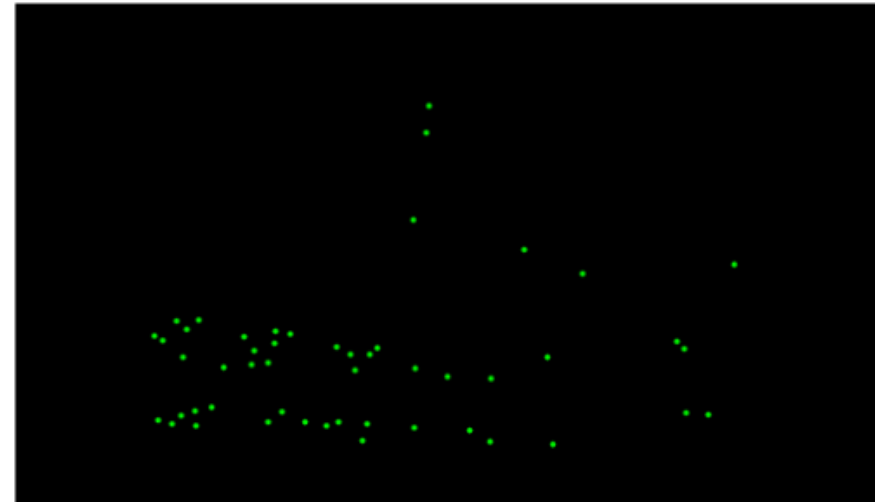
Corner Detection without NMS



Original Image



Corner Detection with NMS



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- Results



HW2: Harris Corner Detection

A. Functions:

- **gaussian_smooth()**: filter images with Gaussian blur.
- **sobel_edge_detection()**: apply the Sobel filters to the blurred images and compute the magnitude and direction of gradient. (You should eliminate weak gradients by proper threshold.)
- **structure_tensor()**: use the gradient magnitude above to compute the structure tensor (second-moment matrix).
- **nms()**: perform non-maximal suppression on the results above along with appropriate threshold for corner detection.

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B. Results:

- a. Original image
 - i. Gaussian smooth results: $\sigma=5$ and kernel size=5 and 10 (**2 images**)
 - ii. Sobel edge detection results
 - (1) magnitude of gradient (Gaussian kernel size=5 and 10) (**2 images**)
 - (2) direction of gradient (Gaussian kernel size=5 and 10) (**2 images**)
(You can choose arbitrary color map to display)
 - iii. Structure tensor + NMS results (Gaussian kernel size=10)
 - (1) window size = 3x3 (**1 image**)
 - (2) window size = 30x30 (**1 image**)
- b. Final results of rotating (by 30°) original images (**1 image**)
- c. Final results of scaling (to 0.5x) original images (**1 image**)

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C. Report:

- a. Discuss the results of blurred images and detected edges between different kernel sizes of Gaussian filter.
- b. Discuss the difference between 3x3 and 30x30 window sizes of structure tensor.
- c. Discuss the effect of non-maximal suppression.
- d. Discuss the results of rotated and scaled image. Is Harris detector rotation-invariant or scale-invariant? Explain the reason.

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D. Notice:

- a. You should **NOT** use any functions which can get the result directly in each steps.
(`cv2.Sobel`, `cv2.Laplacian`, `cv2.cornerHarris`, `skimg.feature.local_binary_pattern`, etc.)
- b. Your code should display and output image results mentioned above.
- c. You should provide a **README** file about your execution instructions.

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A. Functions:

gaussian_smooth(): filter images with Gaussian blur.

```
from scipy import ndimage

def gaussian_smooth(size, sigma=1):
    #####
    # TODO:                                     #
    #   Perform the Gaussian Smoothing         #
    #   Input: window size, sigma              #
    #   Output: smoothing image                #
    #####

    #####
    #                                     #
    #                               End of your code          #
    #                                     #
    #####
    return img

from scipy.ndimage.filters import convolve
img_filtered_K5 = convolve(img_Gray, gaussian_smooth(size=5,sigma=5))
img_filtered_K10 = convolve(img_Gray, gaussian_smooth(size=10,sigma=5))
```

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A. Functions:

sobel_edge_detection(): apply the Sobel filters to the blurred images and compute the magnitude and direction of gradient.

```
def sobel_edge_detection(im):  
    #####  
    # TODO:                                     #  
    #   Perform the sobel edge detection         #  
    #   Input: image after smoothing            #  
    #   Output: the magnitude and direction of gradient #  
    #####  
  
    #####  
    #                                           #  
    #                               End of your code #  
    #                                           #  
    #####  
    return (gradient_magnitude, gradient_direction)
```

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A. Functions:

structure_tensor(): use the gradient magnitude above to compute the structure tensor (second-moment matrix).

```
def structure_tensor(gradient_magnitude, gradient_direction, k):
    #####
    # TODO:                                     #
    #   Perform the cornermess response         #
    #   Input: gradient_magnitude, gradient_direction   #
    #   Output: second-moment matrix of Structure Tensor   #
    #####

    #####
    #                                     #
    #                               End of your code       #
    #                                     #
    #####
    return StructureTensor
```

$$M = \sum_{x,y} w(x,y) \begin{bmatrix} I_x^2 & I_x I_y \\ I_x I_y & I_y^2 \end{bmatrix} = \begin{bmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{bmatrix}$$

$$\det M = \lambda_1 \lambda_2$$

$$\text{trace } M = \lambda_1 + \lambda_2$$

HW2: Harris Corner Detection

A. Functions:

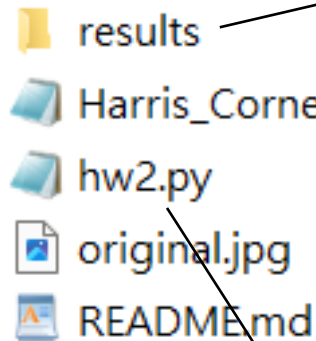
nms(): perform non-maximal suppression on the results above along with appropriate threshold for corner detection.

```
def NMS(harrisim>window_size=30,threshold=0.1):
    #####
    # TODO:                                     #
    #   Perform the Non-Maximum Suppression      #
    #   Input: Structure Tensor, window size, threshold  #
    #   Output: filtered coordinators             #
    #####

    #####
    #                                     #
    #                               End of your code      #
    #                                     #
    #####
    return filtered_coords
```

HW2: Harris Corner Detection

- **Submit File Format:**



- Final results of rotating
- Final results of scaling
- Gaussian smooth results
- Sobel edge detection results
- Structure tensor + NMS results

A. Functions:

- **gaussian_smooth()**: filter images with Gaussian blur.
- **sobel_edge_detection()**: apply the Sobel filters to the blurred images and compute the magnitude and direction of gradient. (You should eliminate weak gradients by proper threshold.)
- **structure_tensor()**: use the gradient magnitude above to compute the structure tensor (second-moment matrix).
- **nms()**: perform non-maximal suppression on the results above along with appropriate threshold for corner detection.

To execute the program and generate images, enter the following command:

```
python hw2.py
```

HW2: Harris Corner Detection

- **ReadMe Example:** <https://github.com/Rossi-Laboratory/Course-Lectures/blob/main/Computer%20Vision/Assignment2/README.md>

Homework 2 Instructions

1. Required Libraries:

The following libraries will be used in this assignment:

`os`, `cv2`, `numpy`, `matplotlib`, `scipy`

Notice: `cv2` should only be used for image conversion (e.g., `BGR2RGB`, `BGR2GRAY`). It **must not** be used directly for the required code in this assignment (e.g., `cv2.Sobel`, `cv2.Laplacian`, `cv2.cornerHarris`).

2. Python Files:

- `hw2.py` (Main execution file)
- `Harris_Corner_Detection.py` (Function file)

3. Running the Program:

To execute the program and generate images, enter the following command:

```
python hw2.py
```



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Q & A



HW2_Introduction.pdf

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